

Matrix Inversion Identities

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Summary

Two simple matrix identities are derived, these are then used to get expressions for the inverse of $(A + BCD)$. The expressions are variously known as the 'Matrix Inversion Lemma' or 'Sherman-Morrison-Woodbury Identity'.

The derivation in these slides is taken from Henderson and Searle [1]. An alternative derivation, leading to a generalised expression, can be found in Tylavsky and Sohie [2].

Two special case results are mentioned, as they are useful in relating the Kalman-gain form and Information form of the Kalman Filter.

Identity 1

$$\begin{aligned}(I + P)^{-1} &= (I + P)^{-1}(I + P - P) \\ &= I - (I + P)^{-1}P\end{aligned}\tag{1}$$

Identity 2

$$\begin{aligned}P + PQP &= P(I + QP) = (I + PQ)P \\(I + PQ)^{-1}P &= P(I + QP)^{-1}\end{aligned}\tag{2}$$

Matrix Inversion Lemma - step 1

For invertible A, but general (possibly rectangular) B,C, and D:

$$\begin{aligned}(A + BCD)^{-1} &= \left(A \left[I + A^{-1}BCD \right] \right)^{-1} \\&= \left[I + A^{-1}BCD \right]^{-1} A^{-1} \\&= \left[I - \left(I + A^{-1}BCD \right)^{-1} A^{-1}BCD \right] A^{-1} \quad \text{Using (1)} \\&= A^{-1} - \left(I + A^{-1}BCD \right)^{-1} A^{-1}BCDA^{-1}\end{aligned}$$

Matrix Inversion Lemma - step 2

Repeatedly using (2) in sequence now produces:

$$(A + BCD)^{-1} = A^{-1} - (I + A^{-1}BCD)^{-1}A^{-1}BCDA^{-1} \quad (3)$$

$$= A^{-1} - A^{-1}(I + BCDA^{-1})^{-1}BCDA^{-1} \quad (4)$$

$$= A^{-1} - A^{-1}B(I + CDA^{-1}B)^{-1}CDA^{-1} \quad (5)$$

$$= A^{-1} - A^{-1}BC(I + DA^{-1}BC)^{-1}DA^{-1} \quad (6)$$

$$= A^{-1} - A^{-1}BCD(I + A^{-1}BCD)^{-1}A^{-1} \quad (7)$$

$$= A^{-1} - A^{-1}BCDA^{-1}(I + BCDA^{-1})^{-1} \quad (8)$$

(note that the order ABCD is maintained, ignoring the other parts of the expressions)

Matrix Inversion Lemma - special case

If C is also invertible, from (5):

$$\begin{aligned}(A + BCD)^{-1} &= A^{-1} - A^{-1}B(I + CDA^{-1}B)^{-1}CDA^{-1} \\ &= A^{-1} - A^{-1}B(C^{-1} + DA^{-1}B)^{-1}DA^{-1} \quad (9)\end{aligned}$$

which is a commonly used variant (for example applicable to the Kalman Filter covariance, in the 'correction' step of the filter).

Another related special case

A very similar use of (2) gives:

$$\begin{aligned}(A + BCD)^{-1}BC &= A^{-1}(I + BCDA^{-1})^{-1}BC \\ &= A^{-1}B(I + CDA^{-1}B)^{-1}C\end{aligned}$$

and for invertible C: (10)

$$= A^{-1}B(C^{-1} + DA^{-1}B)^{-1} \quad (11)$$

which is useful in converting between Kalman-gain and Information forms of the Kalman Filter state-estimate 'correction' step.

Bibliography



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